

FAT 1-DAY AG

Project: SPRI

Item	Time Window	Duration	Milestone / Agenda Item
1	09:00 - 09:15	15 min	Client Arrival & Welcome
2	09:15 - 09:30	15 min	Detail on FAT (Kick-off Briefing)
3	09:30 - 10:45	75 min	Station 1: C1 & C2 Core Audit
4	10:45 - 12:00	75 min	Station 2: C1 & C2 Analog Loops
5	12:00 - 13:00	60 min	Lunch Break & Informal Discussion
6	13:00 - 14:15	75 min	Station 3: C3 (AO) & C4 (P&F IS Barrier)
7	14:15 - 15:00	45 min	Station 4: SCADA & Safety Interlocks
8	15:00 - 15:30	30 min	Coffee Break & Data Consolidation
9	15:30 - 16:30	60 min	Summary Meeting & Comments
10	16:30 - 17:00	30 min	FAT Sign-off & Departure

AGENDA & TIMETABLE — FOCUS ON CHASSIS C1, C2, C3, C4

NT 18K Spray Dryer | Target: MCP-01 (C1, C2, C3, C4) | Ref: KAL-FAT-AGN-2608-C1C4

Key Activities (Focus on Chassis C1–C4)

Client reception, visitor badge, safety induction, ESD protection check before entering MCP-01 bay.

15-Min Kick-off on C1-C4: 4-Chassis DLR ring architecture, test boundary, acceptance criteria, Cat A vs B punch rules.

C1 Master CPU (Slot 0: 1756-L950TPSXT), Cisco 9300 HSRP & DLR ring break test (<3ms), C1 & C2 DI loop toggle (1756-IB32) & DO relay forcing (1756-OB32 -> TBRL -> TB-0VDC).

C1 (C1S10-S11) & C2 (C2S9-S12) 4-20mA calibration audit (Fluke 789 5-point test), verify differential loops (IN-x / i RTN-x) on 1756-IF16, zero crosstalk check.

Executive hot luncheon served in private dining room. Mid-day review of morning C1/C2 test logs.

Chassis C3 modulating valve AO (1756-OF8: 4-20mA) simulation; Chassis C4 dedicated Hazardous Area RIO with Pepperl+Fuchs K-System Zener barrier audit & wire break alarm.

FactoryTalk View SE SCADA integration for C1-C4 tags, faceplates & alarms. Process safety trips: Chamber high-temp trip, burner lockout & Master E-Stop de-energization (<50ms).

Afternoon coffee break, pastries and refreshments. Engineering team consolidates completed C1-C4 test sheets into preliminary Punch List.

Presentation of C1-C4 test results & 100% pass rate. Detailed review of Client Comments. Categorize Punch List (Cat A vs B), assign action owners & due dates.

Formal signing of FAT Certificate of Acceptance for MCP-01 (C1-C4). Review packing protection, dispatch plan & SAT alignment. Departure.



Lead Role	Location / Bay
Safety Lead / Host	Lobby / Conf Room A
Project Manager / Lead Automation	Main Conference Room
Lead PLC Eng / Client Inspector	Test Bay 1 (MCP-01: C1/C2)
Instrumentation Lead / Client Rep	Test Bay 1 (MCP-01: C1/C2)
All Participants	Executive Dining Lounge
Instrumentation & IS Lead / Client	Test Bay 2 (MCP-01: C3/C4)
SCADA Lead / Process Eng	SCADA Control Station
All Participants (Eng team consolidates)	Hospitality Lounge / Office
PM, Lead Engineers & Client Team	Main Conference Room
Signatories & Directors	Main Conference Room

CHASSIS C1, C2, C3,

Chassis	Slot Range	Module Types	Channel Count
C1 (Master)	S0: 1756-L950TPSXT S1-S3: 1756-EN4TR	CPU (20MB) + 3x 1Gbps COMM	1x CPU Core 3x Network Links
C1 (Local I/O)	S4-S7: 1756-IB32 S8-S9: 1756-OB32 S10-S11: 1756-IF16	4x DI (32-pt) 2x DO (32-pt) 2x AI (16-pt)	128 DI, 64 DO, 32 AI Channels
C2 (Process)	S0: 1756-EN4TR S1-S5: 1756-IB32	1x DLR COMM 5x DI (32-pt)	160 DI Channels
C2 (Process)	S6-S8: 1756-OB32 S9-S12: 1756-IF16	3x DO (32-pt) 4x AI (16-pt)	96 DO Channels 64 AI Channels
C3 (Extended)	S0: 1756-EN4TR S1-S3: 1756-IB32 S4-S5: 1756-OB32	1x DLR COMM 3x DI (32-pt) 2x DO (32-pt)	96 DI Channels 64 DO Channels
C3 (Extended)	S6-S9: 1756-IF16 S12: 1756-OF8	4x AI (16-pt) 1x AO (8-pt)	64 AI Channels 8 AO Channels
C4 (IS Barrier)	S0: 1756-EN4TR S1-S3: 1756-IB32	1x DLR COMM 3x IS-DI (32-pt)	96 IS DI Channels
C4 (IS Barrier)	S4: 1756-OB32 S5-S6: 1756-IF16	1x IS-DO (32-pt) 2x IS-AI (16-pt)	32 IS DO Channels 32 IS AI Channels

C4 HARDWARE & TESTING SCOPE MATRIX

Primary Field Destinations	Testing Method
Central Area (CA1), DLR Supervisor, Cisco 9300	CPU power-up, DLR ring break test (<3ms), HSRP switch failover, memory health check.
CA1, JB-401, JB-601	Toggle DI contacts via TB-24VDC; force DO to TBRL relays; 5-point AI calibration with Fluke 789.
JB-402, JB-602, JB-607	Simulate dry contacts from field pumps, valves and limit switches; verify Studio 5000 bit status.
JB-402, JB-602, JB-607, CA1	Force DO bits to interposing relays; inject 4-20mA to differential inputs (IN-x & i RTN-x).
JB-606, JB-612, JB-618	Simulate bag filter differential pressure switches, rotary valve limits, and pulse jet outputs.
JB-607, JB-608, JB-612, JB-618	AI temperature/pressure calibration; force AO modulating setpoints (4-20mA) to control valves.
IS-JB-603, 608, IS-JB-612, 618	Namur proximity switch simulation through Pepperl+Fuchs K-System Zener / Galvanic barriers.
IS-JB-603, 608, IS-JB-612, 618	P&F DO isolator solenoid drive; P&F AI barrier transparency test & wire-break fault detection.



Acceptance Standard
100% Green status, 0 packet drop
DI/DO 100% bit toggle, AI error <0.5% FS
DI transition verified, LED on
Relay contacts closed, AI error <0.5% FS
Pulse jet timing verified, DI/DO ok
AO current output error <0.5% FS
P&F barrier healthy, DI bit toggle ok
Line Fault Detection (LFD) alarms ok

FAT PUNCH LIST & ACT

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ION TRACKING LOG — CHASSIS C1, C2, C3, C4

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FAT ATTENDEE & PARTIC

No.	Full Name	Organization / Company
1	Client Project Manager	Ingredion (Thailand) Co., Ltd.
2	Client Lead Automation / E&I	Ingredion (Thailand) Co., Ltd.
3	Client Plant Operations Lead	Ingredion (Thailand) Co., Ltd.
4	Lead System Integrator	Kalasin Engineering Co., Ltd.
5	Lead PLC / SCADA Engineer	Kalasin Engineering Co., Ltd.
6	Electrical & Wiring Lead	Kalasin Engineering Co., Ltd.
7	QA/QC & Safety Officer	xDev Workshop Integration Center

IPANT REGISTER (MCP-01: C1-C4)

Role / Job Title	Contact Phone / Email	Sign-in Signature
Project Engineering Manager	pm@ingredion.com	
Lead Instrumentation & Control Eng	ei.lead@ingredion.com	
Spray Dryer Process Specialist	ops@ingredion.com	
Project Automation Manager	+66 81-xxx-xxxx	
Senior Automation Engineer (C1-C4)	+66 89-xxx-xxxx	
MCP-01 Hardware Integration Lead	+66 86-xxx-xxxx	
QA/QC Compliance Lead	+66 84-xxx-xxxx	

AUTOMATED DI/DO CL

Objective: Verify wiring fr

Loop #	DO Slot & Pin	Interposing Relay Coil Tag	Relay Dry Contact
LPB-001	C1S8 - Pin 1	TBRL-1 (A1/A2)	TBRL-1 (11/14)
LPB-002	C1S8 - Pin 2	TBRL-2 (A1/A2)	TBRL-2 (11/14)
LPB-003	C1S8 - Pin 3	TBRL-3 (A1/A2)	TBRL-3 (11/14)
LPB-004	C1S8 - Pin 4	TBRL-4 (A1/A2)	TBRL-4 (11/14)
LPB-005	C1S8 - Pin 5	TBRL-5 (A1/A2)	TBRL-5 (11/14)
LPB-006	C1S8 - Pin 6	TBRL-6 (A1/A2)	TBRL-6 (11/14)
LPB-007	C1S8 - Pin 7	TBRL-7 (A1/A2)	TBRL-7 (11/14)
LPB-008	C1S8 - Pin 8	TBRL-8 (A1/A2)	TBRL-8 (11/14)
LPB-009	C1S8 - Pin 9	TBRL-9 (A1/A2)	TBRL-9 (11/14)
LPB-010	C1S8 - Pin 10	TBRL-10 (A1/A2)	TBRL-10 (11/14)
LPB-011	C1S8 - Pin 11	TBRL-11 (A1/A2)	TBRL-11 (11/14)
LPB-012	C1S8 - Pin 12	TBRL-12 (A1/A2)	TBRL-12 (11/14)
LPB-013	C1S8 - Pin 13	TBRL-13 (A1/A2)	TBRL-13 (11/14)
LPB-014	C1S8 - Pin 14	TBRL-14 (A1/A2)	TBRL-14 (11/14)
LPB-015	C1S8 - Pin 15	TBRL-15 (A1/A2)	TBRL-15 (11/14)
LPB-016	C1S8 - Pin 16	TBRL-16 (A1/A2)	TBRL-16 (11/14)
LPB-017	C1S8 - Pin 19	TBRL-17 (A1/A2)	TBRL-17 (11/14)
LPB-018	C1S8 - Pin 20	TBRL-18 (A1/A2)	TBRL-18 (11/14)
LPB-019	C1S8 - Pin 21	TBRL-19 (A1/A2)	TBRL-19 (11/14)
LPB-020	C1S8 - Pin 22	TBRL-20 (A1/A2)	TBRL-20 (11/14)
LPB-021	C1S8 - Pin 23	TBRL-21 (A1/A2)	TBRL-21 (11/14)
LPB-022	C1S8 - Pin 24	TBRL-22 (A1/A2)	TBRL-22 (11/14)
LPB-023	C1S8 - Pin 25	TBRL-23 (A1/A2)	TBRL-23 (11/14)
LPB-024	C1S8 - Pin 26	TBRL-24 (A1/A2)	TBRL-24 (11/14)
LPB-025	C1S8 - Pin 27	TBRL-25 (A1/A2)	TBRL-25 (11/14)
LPB-026	C1S8 - Pin 28	TBRL-26 (A1/A2)	TBRL-26 (11/14)
LPB-027	C1S8 - Pin 29	TBRL-27 (A1/A2)	TBRL-27 (11/14)

LPB-028	C1S8 - Pin 30	TBRL-28 (A1/A2)	TBRL-28 (11/14)
LPB-029	C1S8 - Pin 31	TBRL-29 (A1/A2)	TBRL-29 (11/14)
LPB-030	C1S8 - Pin 32	TBRL-30 (A1/A2)	TBRL-30 (11/14)
LPB-031	C1S8 - Pin 33	TBRL-31 (A1/A2)	TBRL-31 (11/14)
LPB-032	C1S8 - Pin 34	TBRL-32 (A1/A2)	TBRL-32 (11/14)

CLOSED-LOOP CROSS-WIRING SIMULATION (SAMPLE LOG)

From PLC DO to Relay Coil (A1/A2), Relay Contacts (11/14), and DI Input Terminals

Temp Loopback Wire Ref	Paired DI Terminal Block	DI Slot & Pin	Switching Latency
JMP-LPB-001	TBDI-1 (Pin 1)	C1S4 - Pin 1	30 ms
JMP-LPB-002	TBDI-2 (Pin 2)	C1S4 - Pin 2	43 ms
JMP-LPB-003	TBDI-3 (Pin 3)	C1S4 - Pin 3	37 ms
JMP-LPB-004	TBDI-4 (Pin 4)	C1S4 - Pin 4	28 ms
JMP-LPB-005	TBDI-5 (Pin 5)	C1S4 - Pin 5	39 ms
JMP-LPB-006	TBDI-6 (Pin 6)	C1S4 - Pin 6	26 ms
JMP-LPB-007	TBDI-7 (Pin 7)	C1S4 - Pin 7	45 ms
JMP-LPB-008	TBDI-8 (Pin 8)	C1S4 - Pin 8	40 ms
JMP-LPB-009	TBDI-9 (Pin 9)	C1S4 - Pin 9	40 ms
JMP-LPB-010	TBDI-10 (Pin 10)	C1S4 - Pin 10	37 ms
JMP-LPB-011	TBDI-11 (Pin 11)	C1S4 - Pin 11	29 ms
JMP-LPB-012	TBDI-12 (Pin 12)	C1S4 - Pin 12	37 ms
JMP-LPB-013	TBDI-13 (Pin 13)	C1S4 - Pin 13	35 ms
JMP-LPB-014	TBDI-14 (Pin 14)	C1S4 - Pin 14	28 ms
JMP-LPB-015	TBDI-15 (Pin 15)	C1S4 - Pin 15	25 ms
JMP-LPB-016	TBDI-16 (Pin 16)	C1S4 - Pin 16	25 ms
JMP-LPB-017	TBDI-17 (Pin 19)	C1S4 - Pin 19	43 ms
JMP-LPB-018	TBDI-18 (Pin 20)	C1S4 - Pin 20	35 ms
JMP-LPB-019	TBDI-19 (Pin 21)	C1S4 - Pin 21	33 ms
JMP-LPB-020	TBDI-20 (Pin 22)	C1S4 - Pin 22	35 ms
JMP-LPB-021	TBDI-21 (Pin 23)	C1S4 - Pin 23	35 ms
JMP-LPB-022	TBDI-22 (Pin 24)	C1S4 - Pin 24	36 ms
JMP-LPB-023	TBDI-23 (Pin 25)	C1S4 - Pin 25	45 ms
JMP-LPB-024	TBDI-24 (Pin 26)	C1S4 - Pin 26	23 ms
JMP-LPB-025	TBDI-25 (Pin 27)	C1S4 - Pin 27	43 ms
JMP-LPB-026	TBDI-26 (Pin 28)	C1S4 - Pin 28	42 ms
JMP-LPB-027	TBDI-27 (Pin 29)	C1S4 - Pin 29	42 ms

JMP-LPB-028	TBDI-28 (Pin 30)	C1S4 - Pin 30	25 ms
JMP-LPB-029	TBDI-29 (Pin 31)	C1S4 - Pin 31	23 ms
JMP-LPB-030	TBDI-30 (Pin 32)	C1S4 - Pin 32	34 ms
JMP-LPB-031	TBDI-31 (Pin 33)	C1S4 - Pin 33	45 ms
JMP-LPB-032	TBDI-32 (Pin 34)	C1S4 - Pin 34	32 ms

[illegible]

0 -> 1 -> 0	PASS
0 -> 1 -> 0	PASS
0 -> 1 -> 0	PASS
0 -> 1 -> 0	PASS
0 -> 1 -> 0	PASS